

Vehicle Diagnostic Assistant

Problem Statement

Modern vehicles are equipped with advanced electronic systems that generate OBD-II (On-Board Diagnostics) fault codes whenever a malfunction is detected.

Although diagnostic scanners can identify these codes, users often struggle to understand their meaning, possible causes, symptoms, and recommended repair procedures.

Technicians and vehicle owners frequently need to search through multiple service manuals, workshop guides, and online resources to diagnose a problem accurately.

This project aims to develop a **Vehicle Diagnostic Assistant** using **Retrieval-Augmented Generation (RAG)** and **Large Language Models (LLMs)**.

The system integrates a comprehensive OBD-II fault code database with vehicle-specific service manuals and maintenance documents to provide accurate diagnostic information.

Users can select their vehicle model and enter a fault code or symptom to receive detailed explanations, potential causes, troubleshooting steps, repair recommendations, and relevant references through an intelligent conversational interface.